

GIS and Modern Agriculture

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Abstract

This paper considers three peer-reviewed case studies that employ GIS in their investigation of modern agricultural practices. The case studies are presented as examples of how GIS is leveraged in modern agriculture, and how farming practices and technologies benefit as a result. One of the most important applications of GIS in modern agriculture is the identification of land that is best for farming. Walter Musakwa uses GIS and Multiple Decision Criteria Analysis (MCDA) to identify land in South Africa that has the highest suitability for agriculture reform. Farms also require access to massive stores of water for irrigation, so the ability to quantify water consumption in a cost-effective way saves money and time. Santiago Castaño et al. investigate a novel means of identifying another variable to consider when building land suitability models for agriculture – methods for accurately identifying agricultural groundwater consumption by using GIS and remote sensing. Farmers benefit from being able to predict soil loss and erosion, something that is especially advantageous to terrace farming. Pijl et al. use photogrammetry and GIS techniques to measure and model soil loss in terraced vineyards during intense bouts of rainfall, and over longer periods of time.

Regardless of the type of farming, agricultural land must be identifiable and physically stable over time, and resource consumption must be predictable. While the application of GIS in these works greatly benefits the increasingly high-tech agriculture industry, it is important to obtain a holistic perspective by analyzing the observed contributions to modern farming practices. By comparing these various applications, a fuller perspective on the benefits and limitations of GIS in this field is obtained.

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Introduction

GIS is a collection of geospatial technologies, information networks, and means of spatial analysis, that has contributed to the evolution of technology over the better half of the last century. GIS has been made widely available through communications advances in satellite technology like GPS, and through widespread access to improved hardware and software. In the past few decades, the agricultural industry has integrated GIS as a technology to advance and improve the efficiency of modern farming practices. Many modern farming machines rely on precise coordinates, and traditional farming practices have become more aware of the insights that GIS enables. Today most types of farming are in some way reliant on GIS, and the applications of geospatial technologies to agriculture diversify and improve every year. GIS may also be used to improve farming efficiency, making it a more quantifiable, sustainable, and equitable practice.

This paper reviews three academic works that apply GIS to improve modern agricultural practices. The first study uses ArcGIS to produce a suitability report that identifies the best areas for agricultural land reform in South Africa. The second study reveals methods for quantifying groundwater requirements for agriculture in Spain using GIS and remote sensing. The authors of the third study model soil erosion of terrace farms in Italy. These applications of GIS are of major importance to modern farming practices because of the global need to produce more food with fewer resources. By using GIS, the efficiency of farming practices may be maximized so that the production of food can be increased. Monitoring the social and environmental costs of farming may also be integrated to visualize the effects on both communities and natural resources.

Works Reviewed

“Identifying land suitable for agricultural land reform using GIS-MCDA in South Africa”

In this study Walter Musakwa uses ArcGIS and MCDA to “develop a strategically located land index (SLLI) to identify land suitable for agricultural land reform” in South Africa (Musakwa, 2017). By identifying input layers, weighting variables which describe landcover use classes, and mapping, Musakwa produces a strong suitability analysis of potential agricultural land to develop the SLLI. The purpose of his analysis is to improve land reform strategies employed by the South African government to maximize agricultural efficiency and crop production. This study answers how GIS can be used to accomplish this and demonstrates that GIS is an effective tool for providing land use specialists with the ability to make informed decisions based on many criteria.

Government land planners focused on the redistribution of land use in South Africa benefit the most from the information Musakwa presents. Researchers working to create suitability analyses of agricultural land also benefit from Musakwa’s methodology, as it demonstrates effective strategies for determining the most suitable land for agriculture by using GIS models. Musakwa integrates GIS and MCDA to combine geospatial information about land use with preference criteria determined by a group of professionals collaborating on the study. The criteria were ordered and weighted, and rules were assigned for combining data in a GIS model. The GIS model then employed “weighted linear combination (WLC) to generate the SLLI”. Agricultural data on land use, management, and reform were retrieved from legal documents and government

Table 1 Agricultural criteria and weighting

Agriculture criteria	Weight	Rank
Normalised difference vegetation index (NDVI)	15	1
Soil PH	13.4	2
Proximity to rivers and dams	12.2	3
Soil texture	10.1	4
Average annual rainfall	10	5
Elevation	8.7	6
Average temperature min	7.2	7
Average temperature max	7	8
Proximity to roads	4.4	9
Proximity to cities and town	3.6	10
Proximity to economic development corridors (EDC)	2.6	11
Groundwater quality	2.2	12
Proximity to railway line	1.5	13
Proximity to strategic infrastructure projects (SIP)	1.3	14
Proximity to mining/deposits	0.8	15
Total	100	

Figure 1 – List of weighted criteria Musakwa employed to derive the SLLI (Musakwa, 2017).

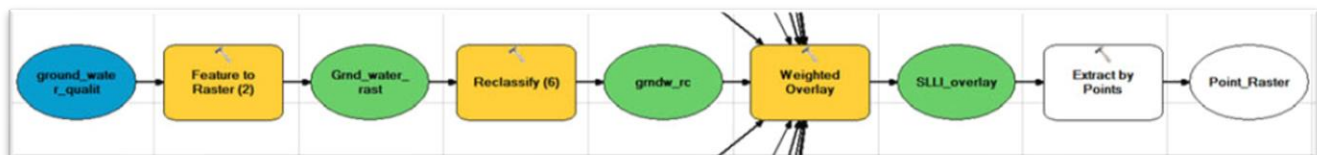


Figure 2 – Zoomed view of a geoprocessing workflow Musakwa used in his model to derive the SLLI (Musakwa, 2017).

sources and were simplified by the team of associated professionals to 15 criteria. These criteria are presented in *Figure 1*. Data on each criterion were retrieved “from the National Geospatial Inspectorate (NGI) and other government departments” and were combined in ArcGIS ModelBuilder to derive the SLLI (Musakwa, 2017) (*Figure 2*).

Musakwa clarifies that the physical and climactic criteria were weighted much higher than proximity data because they have a more significant impact on “development rates of many crops [and] livestock”. Ring buffers were generated around proximity-oriented criteria, and raster layers were generated from the resulting features. Feature layers on the various climactic and environmental criteria were also generated and converted to raster layers. All the raster layers were then reclassified to align with one another and combined into the weighted overlay to produce the

SLLI. The SLLI was then converted to a raster layer of cell size 500 m x 500 m for ease of use and shareability among stakeholders (Musakwa, 2017).

The methods used in this study help to inform public policy on land reform by deriving a reference map that may be used to redistribute land for agriculture in South Africa. The SLLI suitability model allows officials to visualize regions of the country that are the most suitable for agricultural development. This is an important way in which the application of GIS in agriculture helps to inform land planning policy. GIS is used to help provide the South African government with a means of identifying strategically located (highly suitable) land for agricultural use. A concluding focus of the paper identifies strategically located land in regions where it may successfully alleviate food insecurity and inform resettlement planning through sustainable land use. GIS simplifies the “ad-hoc land identification strategies” employed by the South African government by providing the SLLI as a scientific and objective means of land reform planning. This helps to focus attention away from politicized land reuse and toward factual data that was derived and presented by using GIS (Musakwa, 2017).

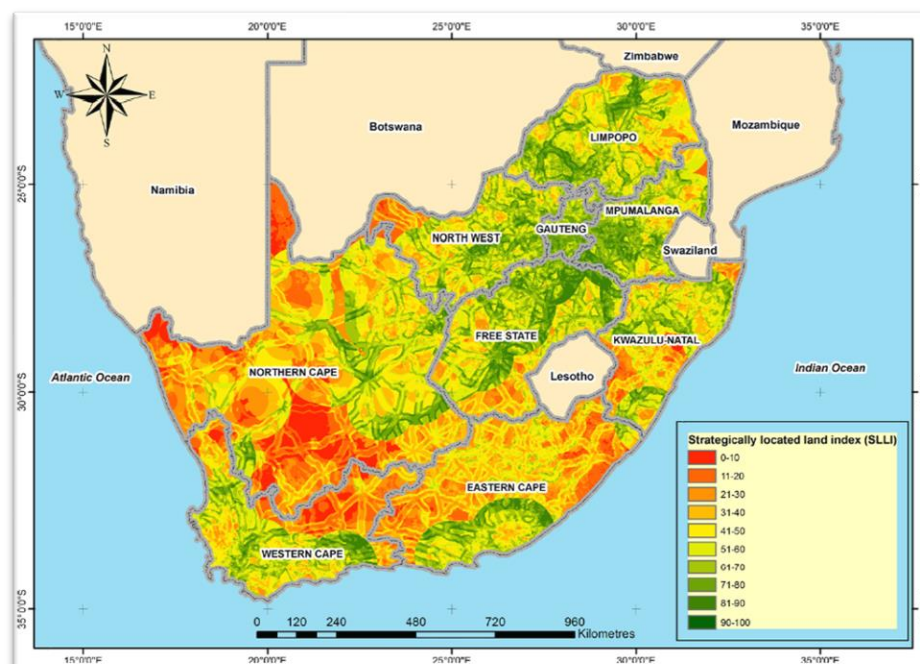


Figure 3 – “Strategically Located Land Index for agricultural land reform (Musakwa, 2017).

“Methodology for Quantifying Groundwater Abstractions for Agriculture via Remote Sensing and GIS”

In this study, Santiago Castaño et al. use remote sensing and GIS techniques to quantify groundwater usage for “agricultural use in semi-arid climates”. By analyzing multispectral satellite images of vegetation coverage, the authors determine irrigation requirements for the observed crops. The aquifer under study in this paper is the Mancha Oriental System (MOS) located in the Jucar River Basin (JRB) in Southeastern Spain, one of the biggest aquifers in the country (7,260 km²). Agricultural irrigation is responsible for 98% of extracted water; the rest supports a population of 275,000 people. The hydrology of the MOS is well known and is used to divide the MOS into six sections to determine hydrogeologic connectedness and fluvial transfer within the aquifer (Castaño et al., 2009).

The purpose of the study is to determine crop irrigation requirements using remotely sensed data. By doing so the “relationships among all elements of the water balance” are visualized (Castaño et al., 2009). This study benefits farmers that rely on aquifers for crop irrigation by enabling them to maximize irrigation strategies and reduce the costs of monitoring water consumption. It also allows farmers to determine how much water their crops require at the beginning of the growing season.

Remotely sensed, multispectral images of the agricultural regions that rely on the MOS were obtained for the years 1982-86, and 1996-2006. The accuracy of vegetation coverage was improved by comparing the evolution of the crops with that of the Normalized Difference Vegetation Index (NDVI), a remote sensing technique for determining live vegetation cover over time. GIS software was used to quantify and organize crop abundance, and to generate theoretical water requirements by seasonal crop type and crop plot location. The information was then used

to produce a hydrologic information system, to visualize the relationships between water abstractions from the MOS and agricultural water consumption (Castaño et al., 2009).

Surface areas of each observed crop type, and theoretical water requirements to yield the observed growth stages of the crops were calculated by using GIS. The irrigation surface areas were determined by using control points to establish a basis for georeferencing polygon layers. A polygon layer for crop plots (1:1000 scale) was overlaid with the crop layer itself (1:100,000 scale) to allow for crop area calculation at the 1:1000 scale.

Irrigation demand for each crop type was obtained from data, published by the Provincial Agricultural Technical Institute of Albacete (ITAP). Water use data between 2001 and 2006 was collected with flow meters on a group of experimental plots in the study area, and the satellite imagery was used to determine the theoretical amount of water required to yield the crops' observed growth stages. This allowed for the empirical determination of correction factors by dividing the effectively used water values into 100 (i.e. how far away the water use at each crop plot was from perfectly effective, each year *Figure 4*). The correction factors were then applied to

Year	Number of plots	Surface (km ²)	IWTV (Mm ³)	IWCU (Mm ³)	Ef (%)	Cc
Spring crops						
2001	20	5.03	1.898	1.610	84.8	1.179
2002	18	5.85	2.682	1.537	57.3	1.751
2003	15	5.99	2.376	1.632	68.7	1.456
2004	17	5.45	1.510	1.178	78.0	1.282
2005	20	6.45	3.565	2.269	63.6	1.572
2006	51	18.55	6.466	4.414	68.3	1.464
Summer crops						
2001	23	6.51	5.698	4.579	82.4	1.214
2002	18	5.74	4.354	3.999	78.5	1.274
2003	14	4.65	3.493	2.940	83.8	1.193
2004	17	5.68	3.720	3.117	84.2	1.188
2005	19	5.61	4.911	3.855	91.9	1.088
2006	27	8.5	5.412	4.462	80.4	1.244
Spring-summer crops						
2001	1	0.59	0.648	0.448	69.1	1.447
2002	1	0.59	0.623	0.448	71.9	1.391
2003	1	0.59	0.612	0.480	78.4	1.276
2004	–	–	–	–	–	–
2005	1	0.46	1.080	0.818	75.7	1.321
2006	5	2.85	2.496	2.442	97.8	1.022

IWTV irrigation water total volume, *IWCU* irrigation water consumptive use, *Ef* irrigation efficiency, *Cc* correction coefficient, (–) no data acquired

Figure 4 – Empirically collected data on water use in the study area from 2001-2006 (Castaño, 2009)

the remotely determined water-use values in the study area between 1982-86 and 1996-2000 to obtain values close to reality, and to observe the evolution of water consumption from the MOS (Figure 5). The results were used to hydrate a geodatabase with the “general hydrological information necessary to perform the integration and interpretation of some results” (Castaño et al., 2009).

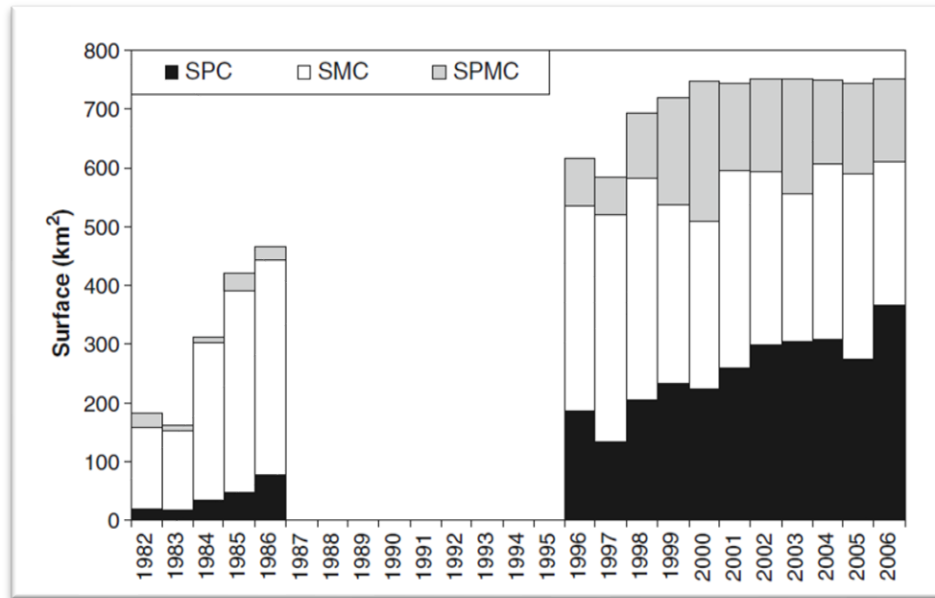


Figure 5 – Evolution of crop areas irrigated by the MOS. (Castaño et al., 2009)

In this application GIS dramatically simplifies and improves traditional methodology for quantifying water requirements for agriculture on a regional level. Until this study there was no cost-effective way to identify water requirements at this scale and resolution. Castaño et al. estimate that this method provides an alternative that is sixty times less expensive than other methods for quantifying extracted groundwater volumes. The technique is cost effective and allows farmers to predict and track water demand and consumption. Given a strong awareness of the hydrological structure of source aquifers and historical data on irrigation efficiency in an area, irrigation requirements of crops may be extrapolated and accurately calculated from multispectral satellite imagery by using GIS (Castaño et al., 2009).

“GIS-based soil erosion modelling under various steep-slope vineyard practices”

This study uses long and short-term analysis methods to determine erosion patterns and quantify soil loss in steep slope agricultural practices at three sites in Northeastern Italy. Terrace farming is a common practice in steep slope environments, and Italy is ideally situated to study soil erosion, given the large amount of annual rainfall there. By using UAV imagery, photogrammetry, and GIS, Anton Pijl et al. produce models to visualize the rate of soil-loss in three different terracing methods in Italy.

The purpose of this study is to produce a long overdue method for determining soil loss potential in steep-slope agricultural environments. The target questions of the study are how rainfall events influence soil loss in terrace farming, and how practical recommendations can be made to improve the efficiency of this method of farming by reducing erosion. Farmers who practice terrace farming will benefit the most from this study by learning how they can improve the design of their farms to minimize soil loss. Those who study the environmental effects of agriculture in these environments will also benefit from the models in this study, as will the governments whose policies they help to inform.

The study implements two models for determining erosion patterns, one for short term soil loss and the other for long term soil loss. The short term Simulated Water Erosion Model (SIMWE) considers “the highly variable data (topography, soil, vegetation)” and is designed for implementation at the crop-plot scale. This model is advantageous for determining erosion patterns that result from single rain events. The longer term Revised Universal Soil Loss Equation model (RUSLE) is a more common model that allows for comparison between the plots observed in this study, and those in other literature that investigates agricultural soil loss (Pijl et al., 2020).

To collect images and produce 3D models of each study site, DJI drones were used with HD cameras mounted on them. From the images point-clouds were generated, and elevation was calibrated from collected reference points. The authors of this study produced an algorithm for creating DTMs from the surface models that resulted from the photo processing. Additional data on physical soil parameters were obtained from an environmental agency in the study region. Other data required for each model were obtained from peer reviewed literature.

GRASS GIS was used to determine steepness and slope aspect coefficients required for modeling the erosion. The results of the SIMWE and RUSLE models were visually analyzed for general erosion patterns (*Figure 6*), and R software was used to compare the different erosion rates observed at each site. Field observations were then compared to the modeled erosion patterns. The results of the RUSLE model did not effectively show accurate locations of erosion and were therefore used as an overall estimate of erosion severity for the sites. The SIMWE model however produced erosion patterns that were consistent with field observations. This model is therefore used to determine “the degree of soil loss that could be linked to the risk of major damages” (Pijl et al., 2020).

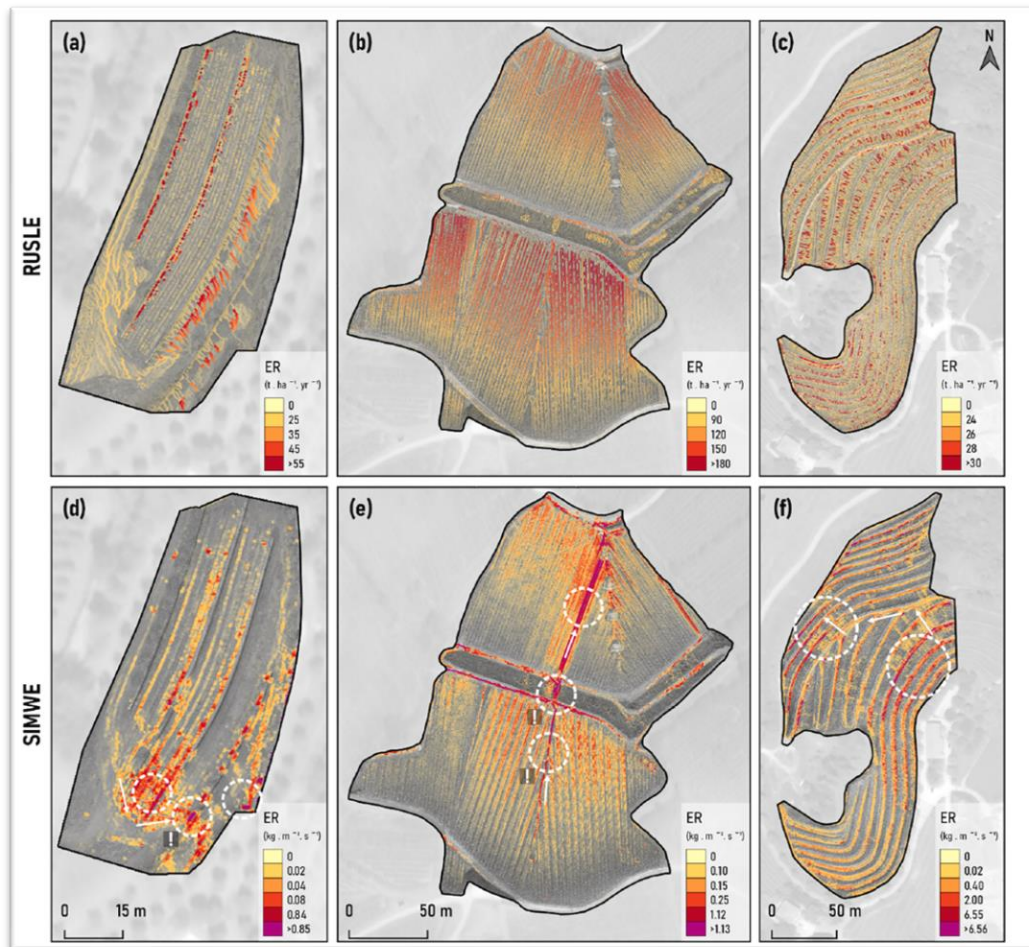


Figure 6 – Erosion rates (ER) for the study areas; long term (RUSLE), short term (SIMWE). (Pijl et al., 2020)

The practices used in this study rely on GIS to determine coefficients that are required for the SIMWE and RUSLE models to produce results that are consistent with field observations. GRASS GIS allowed for the calculation of numerical values associated with the DTMs generated in this study. In this way GIS served as a middle-ground between the obtained imagery and the resulting models. This allows for the complete analysis of the two methods used to model soil erosion, and consequently for the comparison between the models and field observations. GIS improves the methodology in this study by streamlining the analysis of the DTMs produced by the authors. It also simplifies the process of calculating the slopes and their derivatives at every point in the models.

Synthesis and Discussion

Similarities of Methodology

Of the three studies there is some overlap in terms of GIS methodologies used. The agricultural land suitability analysis performed by Musakwa relies among other things on the collection of geophysical data on soil characteristics, information that is heavily required by Pijl et al. to determine erosion potential. Terrain characterization and elevation modeling are also demanded in both studies to produce accurate models and suggestions for improving agricultural land use. GIS is most heavily relied upon for data manipulation and analysis in the works by Musakwa and Castaño et al.

Each study uses GIS to quantify resources that efficient agriculture practices rely on. In Musakwa's case the resource is land, and the distribution of highly suitable land is complicated by socioeconomic and political characteristics of South Africa. Castaño's team is more concerned with the precious resource of water, and how GIS can be used to determine and monitor its availability in the most cost effective and least intrusive way. His analysis is complicated more by the availability of empirical data to determine correction coefficients than it is by social factors. The similarities between the works of Pijl et al. and Castaño et al. are slightly more ambiguous, though both teams rely on their own field studies to qualify their results, something Musakwa did not require. Pijl and Castaño also rely on aerial photography to produce the data they analyze in their respective studies. Overall, GIS is used to simplify data processing in each report, and all authors require it to perform the level of spatial analysis they present.

Differences of Methodology

While there are a host of similarities among these three studies, there are also major differences in terms of how GIS practices are applied. The first major difference is between Musakwa and Castaño's publications; where Musakwa is concerned with determining suitability characteristics of land with a focus on agriculture, Castaño focuses more on integrating remotely sensed data with observed agricultural data by using GIS. Pijl et al. also focus on integrating aerial data with field observations, but terrace farming in Italy is quite different from farming practices in a semi-arid environment like Southeastern Spain. Soil erosion is a much larger concern in Northeastern Italy, especially given the predictions of increased precipitation there in the future. Aquifer depletion is of higher concern in Southeastern Spain than in Northeastern Italy. Pijl and Musakwa's studies also diverge in most ways, particularly in terms of their reliance on GIS to produce informative material that helps to advance modern agriculture.

Musakwa's work involves a purer application of GIS than do the other two studies. After collecting and organizing the right data, his suitability model entirely enables the production of the SLLI. Castaño and Pijl's works on the other hand involve data manipulation and processing outside of a strict GIS environment, though the techniques are related. The results of these works also allow for more decision making at the individual farm level than Musakwa's, though Pijl's results can more easily suggest advantageous changes to farming practices than Castaño's. This is due to the fact that terraces can be redesigned, but groundwater stores cannot be increased.

Effectiveness of Methodology

While they target different stakeholders in the agricultural industry, the methodologies used by the authors of these studies are quite effective. Musakwa's tactic of using ArcGIS ModelBuilder does a great job of simplifying the data layers his team determined to be most relevant. This approach simplifies many criteria into a single model that can be used by governments and policy planners to identify land suitability for agricultural reform. Castaño's team effectively uses GIS to measure surface areas, and they rely on it for processing and integrating their remotely sensed data. They also implement GIS to produce a five-year study to quantify groundwater usage experimentally, and then integrate the results with the theoretical abstractions determined from their remote sensing work to help normalize the observed data. While Pijl's team used GIS the least, they greatly rely on it to simplify the process of determining critical inputs for their erosion models. This is a very effective use of GIS, and it improves accuracy and minimizes confusion when presenting their results to a public audience.

Castaño and Pijl's methods are novel in their approach to using GIS in agriculture. Castaño's team is the first to develop a method for using remotely sensed satellite imagery to determine water requirements for seasonal crops, allowing for accurate calculations of groundwater abstractions from a source aquifer. Pijl's team develops an algorithm for producing DTMs from point-cloud data collected using drones and photogrammetry, for specific use with widely adopted erosion models and techniques. While Musakwa's techniques are not novel, he provides the South African government with the SLLI, a tool they never had before. This may be used to improve the effectiveness of land redistribution practices, helping both to promote efficient farming and social equity.

Overall new information is derived in each study. Musakwa provides the first ever scientifically objective suitability analysis for potential agricultural land in South Africa. Along with data analysis of specific agricultural practices and careful mapping of land-use characteristics, Musakwa's study provides local governments with a powerful tool for agricultural land planning by using GIS. Castaño et al. also provide a totally novel means of quantifying groundwater usage in agriculture. Given a few moderately easy to obtain inputs, researchers can effectively and accurately determine agricultural water usage levels in a cost-effective way. This may help to optimize land-use efficiency in developing parts of the world where technological resources may be limited or difficult to access. Pijl et al. also advance modern agricultural practices in terrace farming by using GIS to fulfill highly detailed erosion models that predict the location and severity of soil loss during intense rain events.

Summary and Conclusion

While these studies provide great insight into a handful of applications of GIS in agriculture, improvements may be suggested to help integrate the results with modern technology and promote access to a wider audience. Musakwa's suitability analysis completely avoids the complex sociopolitical terrain of South Africa and may be improved with the inclusion of data on socioeconomic factors. This would support his concluding remarks on considering the social qualities of land-use in South Africa when determining suitable land for agricultural reform. Layers on wealth and demographics would give great insight into where land may be easiest to repurpose for agriculture. While he does not include these types of layers, his SLLI provides a valuable means for other analysts to include this data in their own work. Musakwa's study would also be improved by describing the limits of the employed WLC, to help disambiguate the accuracy of his results.

The work of Castaño et al. may be improved in a similar way, where additional data layers could be implemented to suggest improvements for the efficiency of groundwater consumption. Crop plots could be symbolized by consumption efficiency to identify which plots use water the most inefficiently. Castaño's study could also benefit from an approach like Musakwa's, to include a suitability analysis for identifying plots which are the easiest to irrigate. This would allow stakeholders to determine which plots are using water the most efficiently, and when compared with the previously suggested layer on crop consumption efficiency, overall efficiency ratings may be observed.

Pijl's team could improve their study by presenting their findings in a 3D ArcGIS scene type view, where users could easily interact with the results themselves. It would also have been helpful to have the modeled results of a handful of different rain event intensities overlaid, to show

more clearly how erosion rate relates to rain severity. Another means of improving their results would be to produce a time-dependent view of erosion rates with a time-slider function, for digital use. This would make it easier to visualize their results and may serve as a framework on which future erosion models could be dynamically reviewed. Their publication could also be enhanced with the inclusion of maps of erosion potential instead of erosion rates, making their results easier to conceptualize.

The reviewed works provide a well-rounded insight into how GIS is used to improve modern agricultural practices in different parts of the world. Through a variety of data collection methods, manipulation techniques, and analytical procedures, farming is made more effective and sustainable. Farmers are able to leverage GIS to predict and monitor geospatial phenomena that have thus far been difficult to manage. While the suggested improvements may make the results of the studies more digestible, the authors do a very good job at applying GIS in an agricultural context, helping to benefit administrative stakeholders, and farmers themselves. They are clear examples of why GIS is becoming a network of processes and technologies that is increasingly relied upon to advance agricultural practices and adapt to demands of the modern world.

Works Cited

- Musakwa, Walter. "Identifying Land Suitable for Agricultural Land Reform using GIS-MCDA in South Africa." *Environment, Development and Sustainability*, vol. 20, no. 5, 2018, pp. 2281-2299. *ProQuest*, <http://proxy.library.jhu.edu/login?url=https://www.proquest.com/docview/2102401914?accountid=11752>, doi:<http://dx.doi.org/10.1007/s10668-017-9989-6>.
- Castaño, Santiago, et al. "Methodology for quantifying groundwater abstractions for agriculture via remote sensing and GIS." *Water Resources Management*, 2010, pp. 795-814. *GeoScienceWorld*, [http://www.springerlink.com/\(0j23p045e4cou0nbbennj355\)/app/home/journal.asp?referrer=parent&backto=linkingpublicationresults,1:103011,1](http://www.springerlink.com/(0j23p045e4cou0nbbennj355)/app/home/journal.asp?referrer=parent&backto=linkingpublicationresults,1:103011,1).
- Pijl, Anton, et al. "GIS-based soil erosion modelling under various steep-slope vineyard practices." *Catena*, vol. 193, article 104604, 2020. *ScienceDirect*, <https://www-sciencedirect-com.proxy1.library.jhu.edu/science/article/pii/S0341816220301545>